Tabular AI & SAP-RPT

Turning powerful tabular models to production-ready predictive capabilities

Partner Enablement

11 August 2026

Large language models

AI for unstructured data

New category of models

Tabular foundation models

AI for structured data

Large language models

AI for unstructured data

Predictions

- Sales Quotation Conversion Rate Prediction
- Sales Fulfillment Delay Prediction
- Intelligent Product Recommendation
- Material Group Prediction
- Goods Receipt/Invoice Receipt Reconciliation
- Return Order Auto-completion
- Payment Anomaly Detection
- Commodity Code Prediction
- Tax Code Prediction
- Mid-term Cashflow Forecasting
- Spend Categorization
- Incoming Payment Delay Prediction
- Intelligent Accruals
- Sale Order Auto-completion
- Recommendation of Destination Bins
- Damage Code Proposal
- Guided System Configuration
- Dynamic Stock Level Optimization
- Advanced Returns Management
- Identification of Failure Modes
- Product Upselling Recommendation
- Project Cost Forecasting
- Lead Time Prediction for Demand-driven Replenishment
- Delivery Delay Prediction
- Credit Risk Prediction
- Estimate Emissions Factor
- Sustainability Taxonomy Classification
- Self-healing Master Data
- Guided System Migration
- Invoice Object Recommendation

Note that these are not live use cases but examples of predictive scenarios

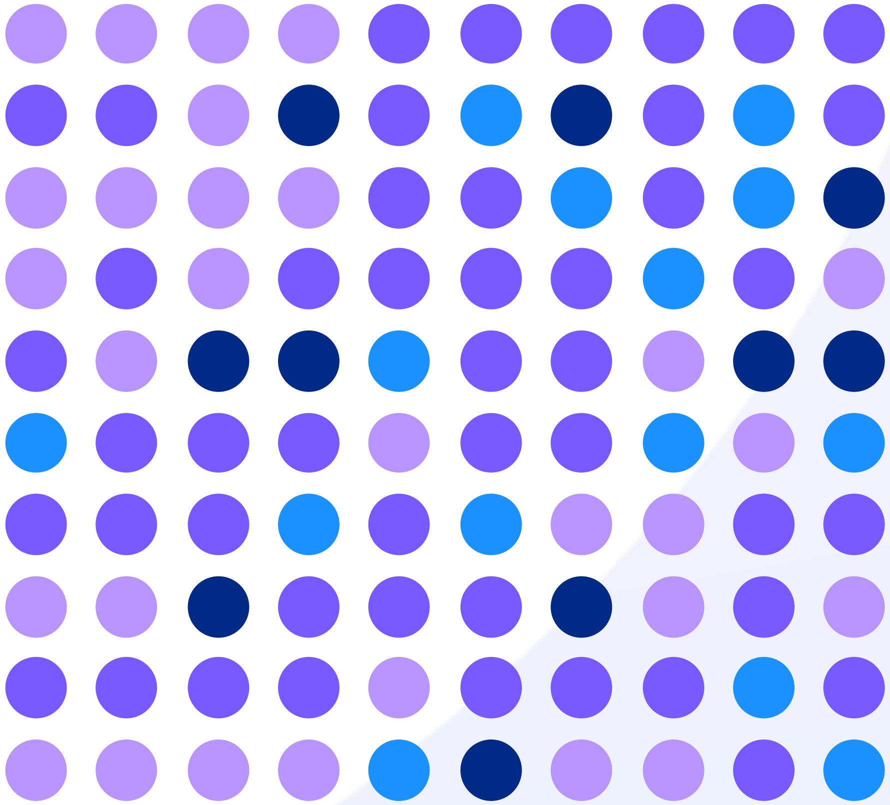
10 Predictive tasks

- Payment Anomaly Detection
- Sales Fulfillment Delay Prediction
- Sales Order Auto-completion
- Return Order Auto-completion
- Intelligent Product Recommendation
- Material Group Prediction
- Invoice Receipt Reconciliation
- Delivery Delay Prediction
- Invoice Object Recommendation
- Dynamic Stock Level Optimization

10 Company codes

- DE01
- BR01
- CA01
- FR01
- EG01
- SG01
- IN01
- NZ01
- MX01
- SA01

100 Models



1

Model

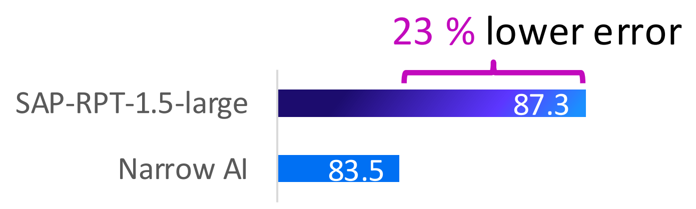
SAP-RPT

Relational Pretrained Transformer

SAP-RPT - 1.5

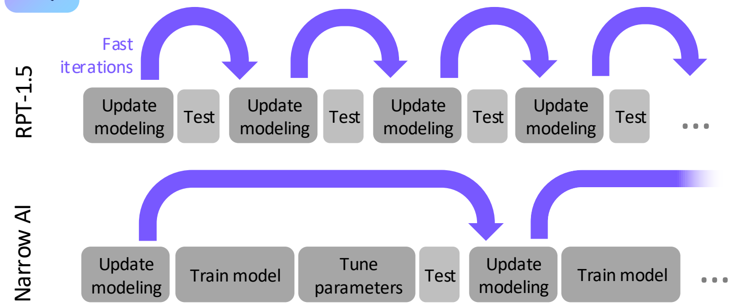
SAP-RPT in a nutshell

More accurate predictions



Average accuracy (↑) across 10+ SAP scenarios, 200+ customer datasets

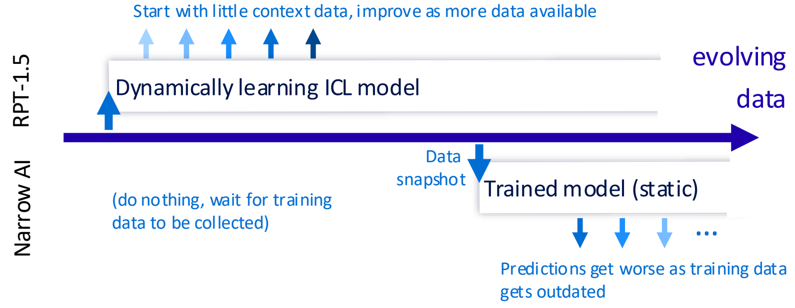
Faster development cycles



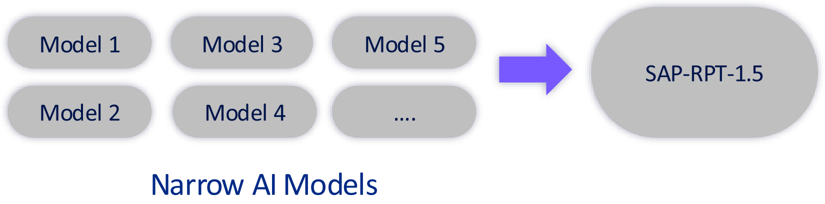
Unlocking
new use cases

Democratizing &
Scaling Tabular AI

Speed-to-value & continuous learning



Kill model sprawl



In-Context Learning for Tabular Data

1

Provide examples and questions together

Input

The context

Invoice number	Line item	Supplier	Price	General ledger account
0001	1	Stark Industries	100	100
0001	2	Stark Industries	200	101
0002	1	Gringotts Bank	4000	100
...	...	...	...	...

The query /

Invoice number	Line item	Supplier	Price	General ledger account
0003	1	Soylent Corp	100	???
0003	2	Soylent Corp	200	???

Engine

2

Enable the model to understand the task from the context during a single, real-time forward pass

Output

Invoice number	Line item	Supplier	Price	General ledger account
0003	1	Soylent Corp	500	100
0003	2	Soylent Corp	1000	101

3

Get instant, task-specific predictions without any model retraining

SAP-RPT-1.5

Now generally available on gen AI hub

- **Explainability**
  - Classical column-wise explanations (“feature importances”)
  - Novel context-row-based explanations (“explain by example”)
- **Improved performance** for large-scale data:
  - ~4.3pp – 5.9pp accuracy gains for high-cardinality use cases
  - ~7.5pp – 11.9pp F1 Macro score gains for large SAP datasets
- Context management & unlimited context via Tabular Orchestration *(GA planned for August 2026)*


```
"predictions": [
  {
    "COSTCENTER": [
      {
        "prediction": "Office Furniture",
        "confidence": 0.96,
        "confidence_interval": null
      }
    ],
    "ID": "35"
  }
],
"explanations": {
  "top_column_scores": [
    {"DESCRIPTION": 0.523, "AMOUNT": 0.234
    {"AMOUNT": 0.612, "DESCRIPTION": 0.188
  ],
  "top_relevant_context_rows": [
    [3, 4, 2],
    [4, 3, 2]
  ]
},
```


Detailed benchmarking results are published in our [model card](#)
More information can be found on our [Help Portal](#) page

SAP-RPT-1.5 Variants

SAP-RPT-1.5

Optimized for **speed**

Available via RPT playground

SAP-RPT-1.5-Large

Optimized for **accuracy**

No changes to pricing and model specifications.

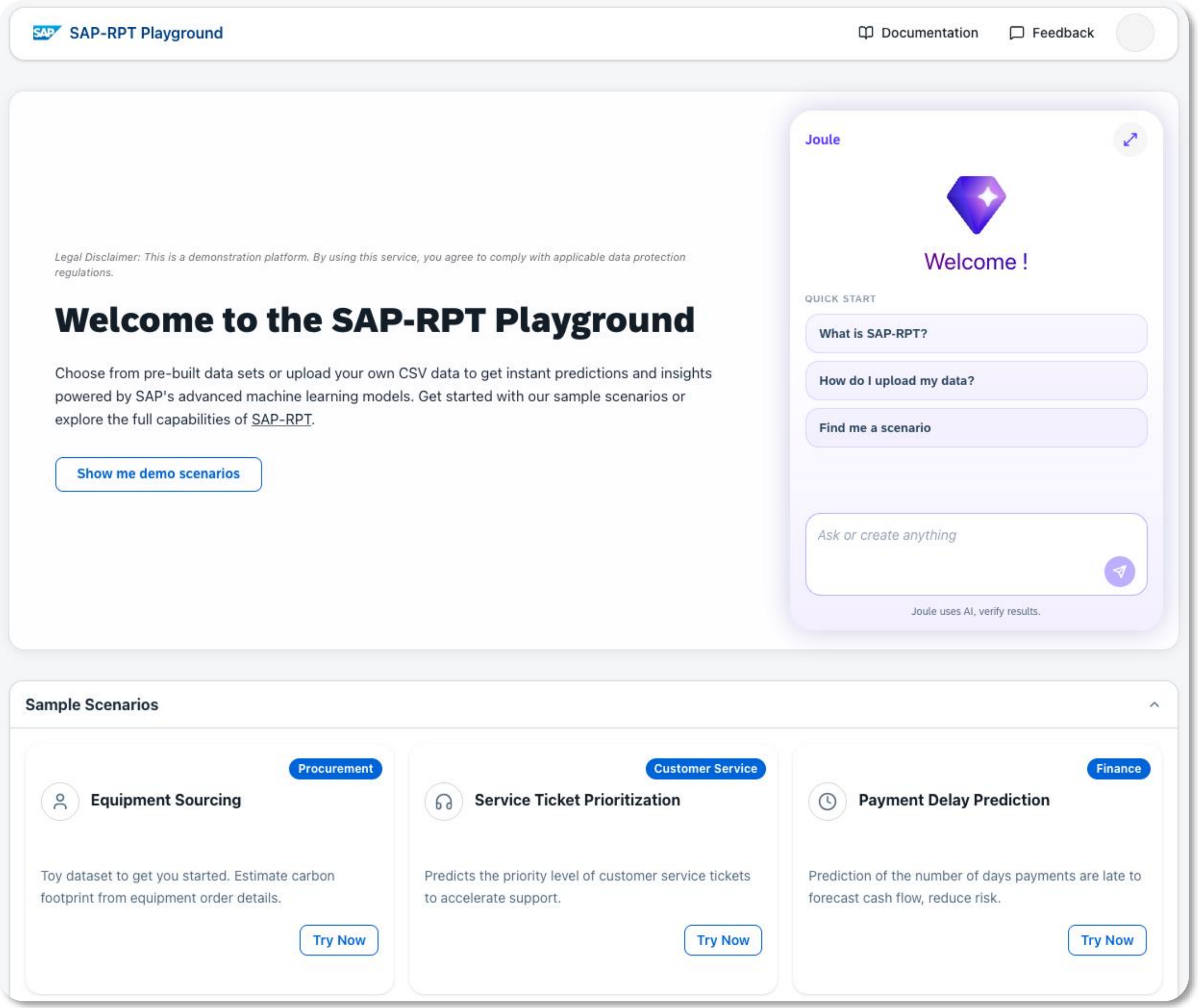
RPT-1 and RPT-1.5 will be priced identically.

See SAP Note on model availability and cost. Use the AI Core Calculator for cost estimation.

SAP-RPT External Playground

SAP-RPT Playground

ConTextTab Paper



Tabular AI in SAP

RPT models are powerful model engines but use cases and business processes need more than just a component

They need a system that removes the complexity and handles the **heavy lifting** of turning a model into a reliable, governed business process

We envision Tabular AI to be the **vehicle** that supports the use of our core model engines

Key Objective 1

Unified interface for all (existing and future) predictive models

Key Objective 2

Extensive tooling to support easier use of predictive models

Key Objective 3

E2E build-test-deploy workflow to build and run any predictive use cases

Tabular Orchestration

- Available on gen AI hub as a new orchestration workflow for tabular foundation models (TFMs)
  - Default and production-ready integration pathway for RPT models and new TFMs
- Main mode of consumption: **API**

Key Features

- Centralized Context Management – connect your HANA table and the workflow can manage the data retrieval
- Automated Context Selection – select from available context selection strategies

▶ GA workflow with targeted context selection targeted for **Aug 2026**

Why use Tabular Orchestration?

Automated Context Management

Workflow handles the ingestion of context data and automates the selection of relevant context rows

Better predictions from using our **latest targeted context retrieval approach**, which gives an uplift over baseline context selection methods for RPT models

Compatibility

Tabular Orchestration is the **harmonizing layer** across existing and new tabular foundation models

Switch easily between TFMs via a common interface

Getting Started with Tabular Orchestration

Design-Time Steps

(Done once per use case)

1. Set up a data destination
2. Register your tabular artifact
3. Create your AI use case scenario
4. Create a virtual deployment of Tabular Orchestration

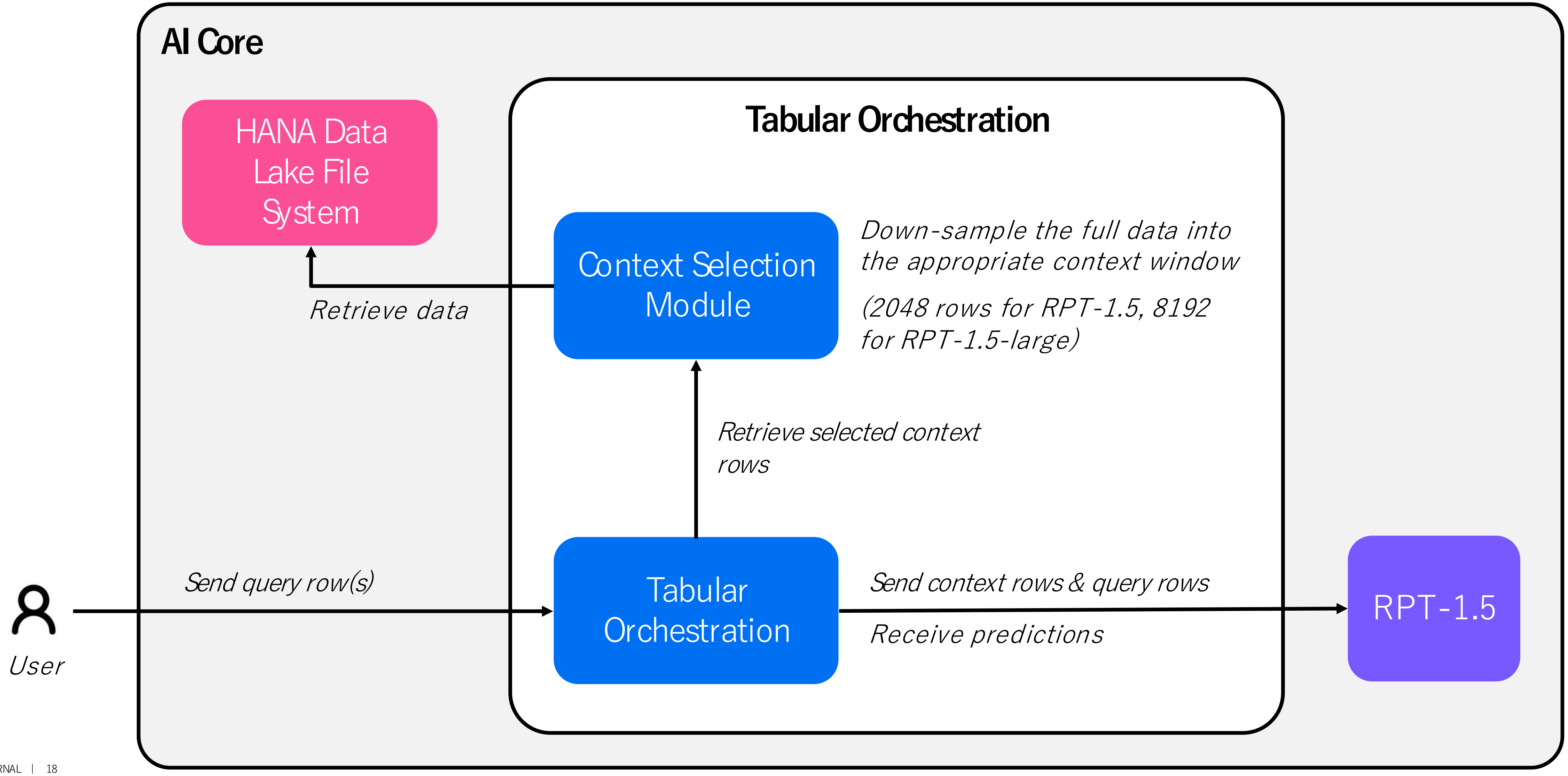
   (Done once per service key, i.e., multiple scenarios can use the same virtual deployment)

Run-Time Steps

(Call whenever you require predictions)

1. Call the Tabular Orchestration workflow with your defined AI use case scenario

How It Works


AI Core
HANA Data Lake File System
Tabular Orchestration
Context Selection Module
Down-sample the full data into the appropriate context window
(2048 rows for RPT-1.5, 8192 for RPT-1.5-large)
Retrieve data
Retrieve selected context rows
Send query row(s)
User
Tabular Orchestration
Send context rows & query rows
Receive predictions
RPT-1.5

Let’s get hands-on!

Hands-On: Introduction

In the exercise, we will use **Predicting Employee Flight Risk in HR** as an example use case.

Through the exercise, you will get to:

- Run the steps of setting up the Tabular Orchestration workflow for the example use case
  - Set up a data destination
  - Register your tabular artifact
  - Create your AI use case scenario
  - Create a virtual deployment for the workflow
- Run the workflow to get predictions from SAP-RPT-1.5 for the example use case

Hands-On: Getting Started

Head to the [WorkZone page]() and download the zip file to get started.

The zip file contains:

- Raw Datasets
  - `Person_Employment_Info.csv`
  - `Succession_Export.csv`
  - `User_Directory.csv`
- Python notebook for the hands-on exercise (`Hands_On_Exercise.ipynb`)
- Credentials folder with HDLFS client certificate and keys as well as AI Core service key
  - *Note that credentials will only be available for 24 hours*
- Bruno collection for all API requests required
  - *Optional for the Hands-on. We provided a Bruno API collection in case you would like to run any individual API calls.*

Hands-On: Getting Started

Extract the zip file.

Open the extracted folder in your preferred code editor.

Setup a local environment using the provided `requirements.txt`. If you are new, you can follow the instructions here.

Start the local environment.

Open the Python notebook.

Run the code blocks sequentially.

Hands-On: Context Selection Strategies in Tabular Orchestration

This can be toggled via the `contextSelectionConfig.strategy` parameter, which is an optional parameter

Supported values include:

- `auto`: workflow will use heuristic by default, fallback to random if there is an error (e.g., timeout)
- `heuristic`: fetch context rows semantically closest to each query row, fails if there is an error (e.g., timeout)
- `random`: sample context rows at random (fast, good baseline)
- If no value was provided, the workflow will use random

Hands-On: Important Notes

- For this exercise
  - All participants will be sharing a common HDLFS instance on our Canary landscape
- Make sure that all names used when creating AI Core components are in lowercase alphanumeric with hyphens only
- RPT-1.5 is already generally available for you to try out.
  - Refer to the help portal guide for more details.
- Tabular Orchestration will only be generally available from early September 2026.
  - The exercise was created using the sap-internal plan. You should create an AI Core under the **extended** plan when you are setting it up on your own.
  - You will be able to follow the same steps in your own setup once it is available.
  - The workflow currently supports up to 32 query rows per call.
  - The workflow currently does not yet return the explanations from RPT-1.5. This will be supported at the time of GA.

Hands-On: Extension

As an extension, you can adjust the provided code for your own use case.

Duplicate the provided notebook. Use your own dataset(s) and edit the data preparation logic as required. You can go on to upload your context data to the common HDLFS instance.

Make sure to not upload sensitive data. Ensure that there are no PIIs and data have been properly masked / anonymized.

To get a feel of the full benefit of the targeted context selection strategy, we recommend you using datasets that are much larger than the max context windows for the model variants

- 2048 context rows for RPT-1.5
- 8192 context rows for RPT-1.5-large

Q&As

Thank you.

Contact information:

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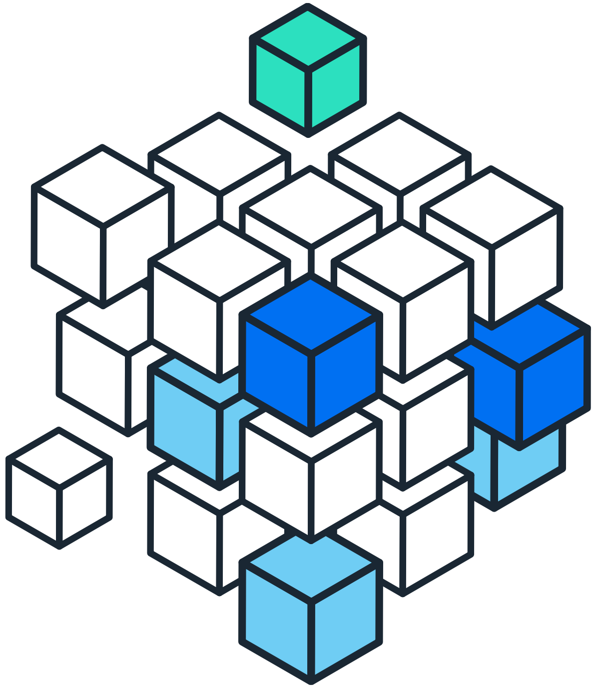
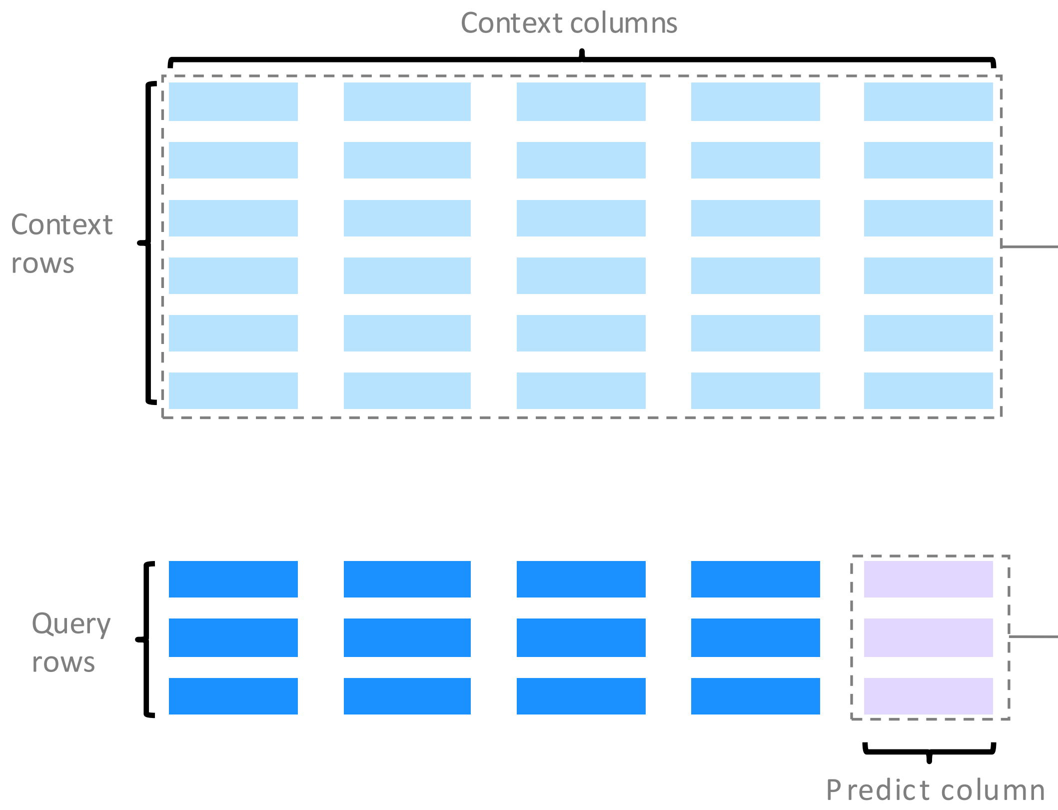


How RPT Pricing Works



Input Cells

# of context rows * # of context columns

Predict Cells

# of query rows * # of predict columns

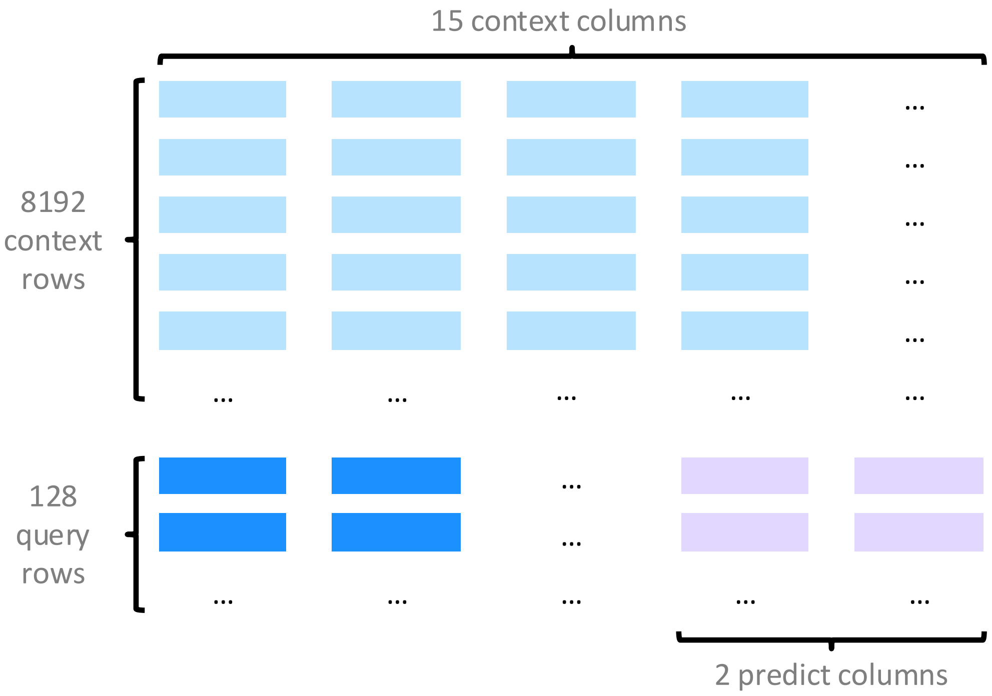
Cost per API = Cost of input cells + Cost of predict cells

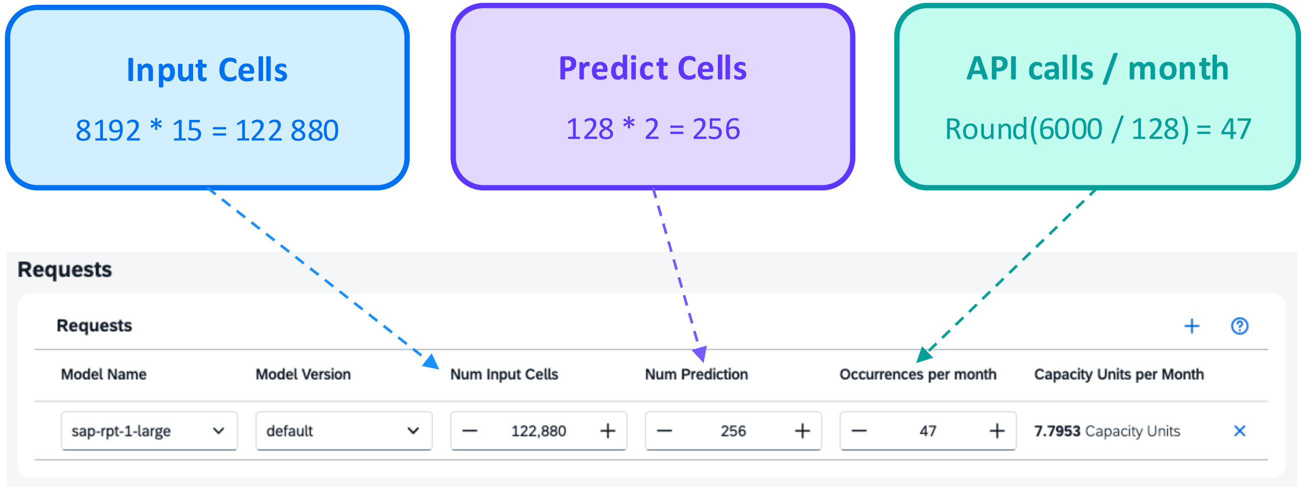
See [SAP Note] on model availability and cost. Use the [AI Core Calculator] for cost estimation.

How RPT Pricing Works

Example Use Case: Sales Order Completion

- 8192 historical records with 15 features
- 128 query rows per API, each with 2 predict cells
- Assume 6000 sales order per month
- Using SAP-RPT-1.5-Large





Setting Up AI Core on BTP

Entitlements

Ensure that your Global Account admin has granted the following entitlements to your subaccount:

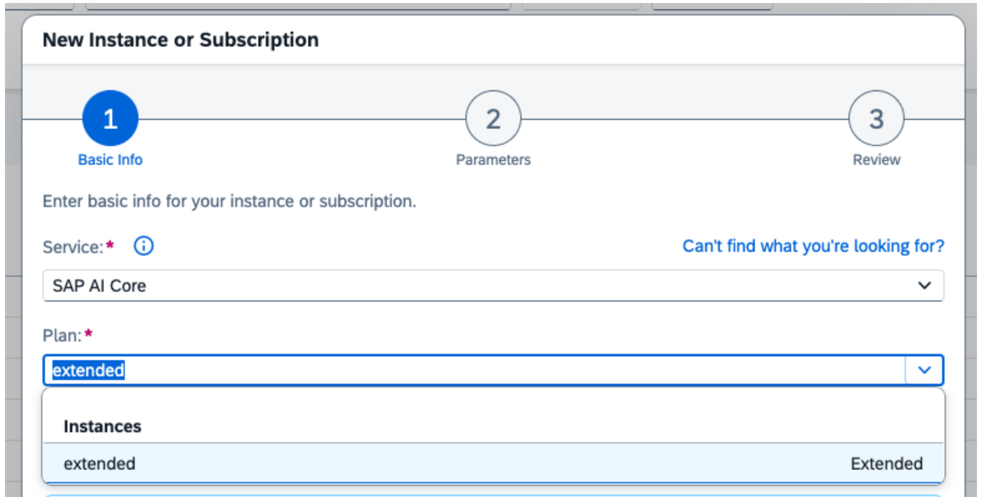
- ai-core
- ai-launchpad

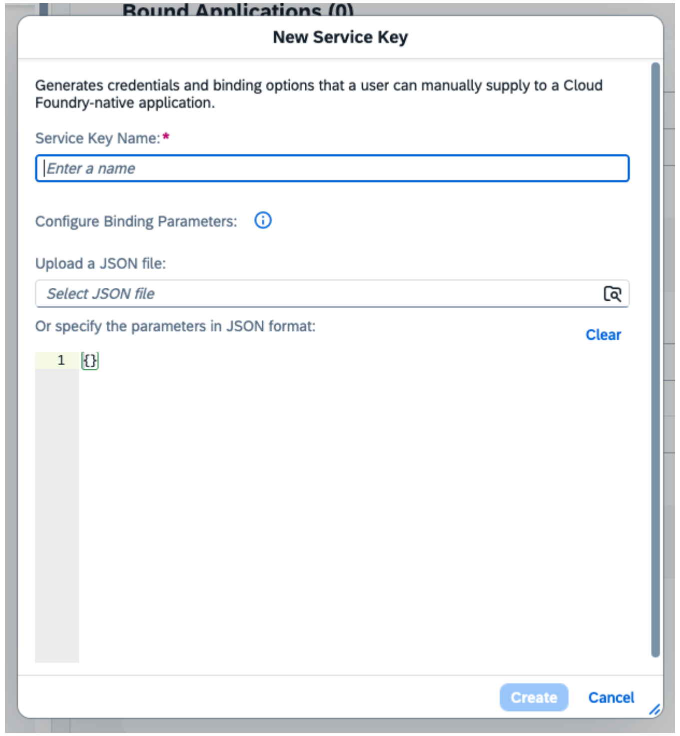
Quota is dependent on your specific needs

Setting up AI Core

Make sure to create an AI Core instance under the **extended** plan

Once the instance is created, under the **Service Key** section, click **Create** to create a service key. Give it any name you'd like and click **Create**.





Obtaining AI Core Service Key

Once the service key is created, you will see that it is JSON with several key-value pairs

Note down the following:

- `url`, `clientid` and `clientsecret` – required to obtain the bearer token
- `AI_API_URL` – the base URL for any calls to API AI Core

The bearer token is needed for every API call to AI Core

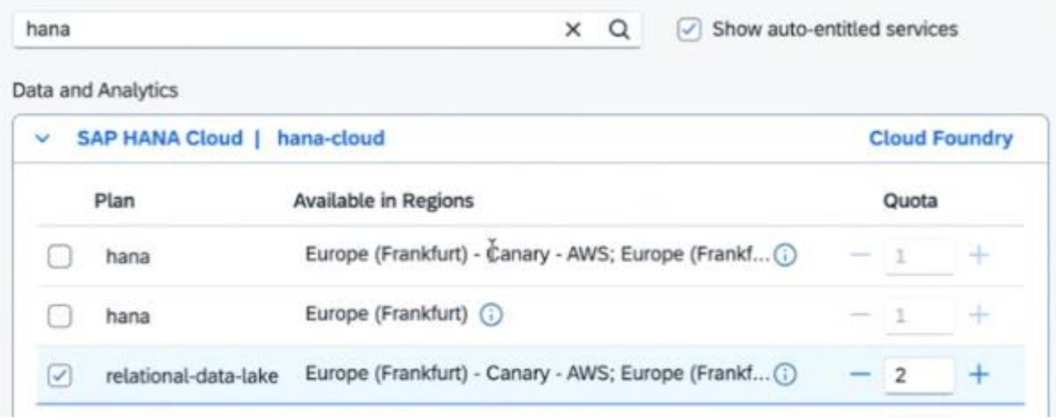
Setting Up HDLFS

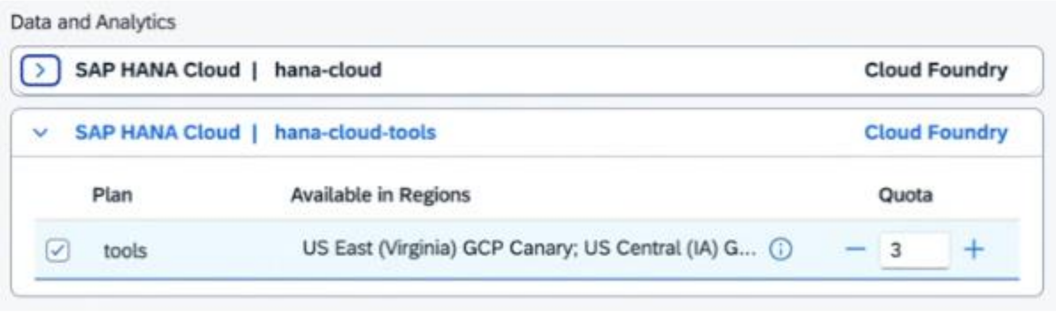
Entitlements

Ensure that your Global Account admin has granted the following entitlements to your subaccount

- hana-cloud > relational-data-lake
- hana-cloud-tools > tools

Quota is dependent on your specific needs

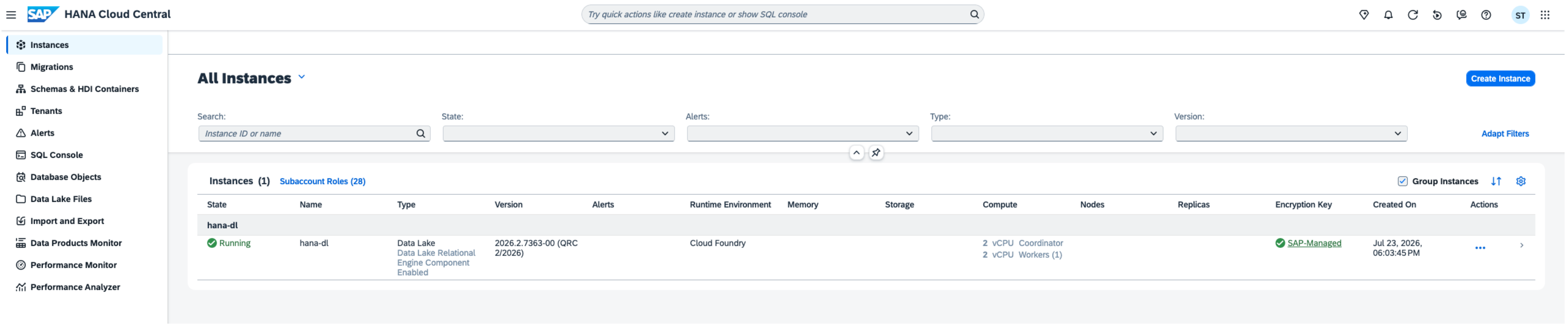




Setting up HDLFS

Create an SAP HANA Cloud application and a HANA instance under **relational-data-lake** plan

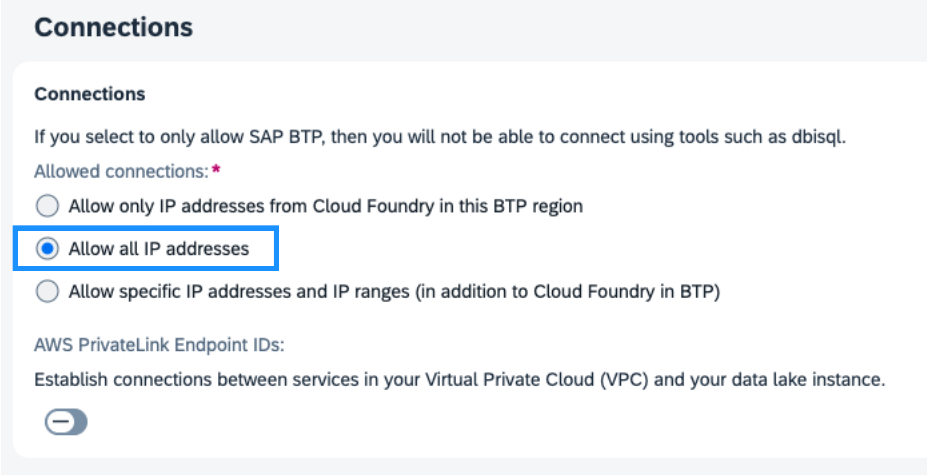
Once created, access the dashboard and you should already see 1 instance running



Setting up HDLFS

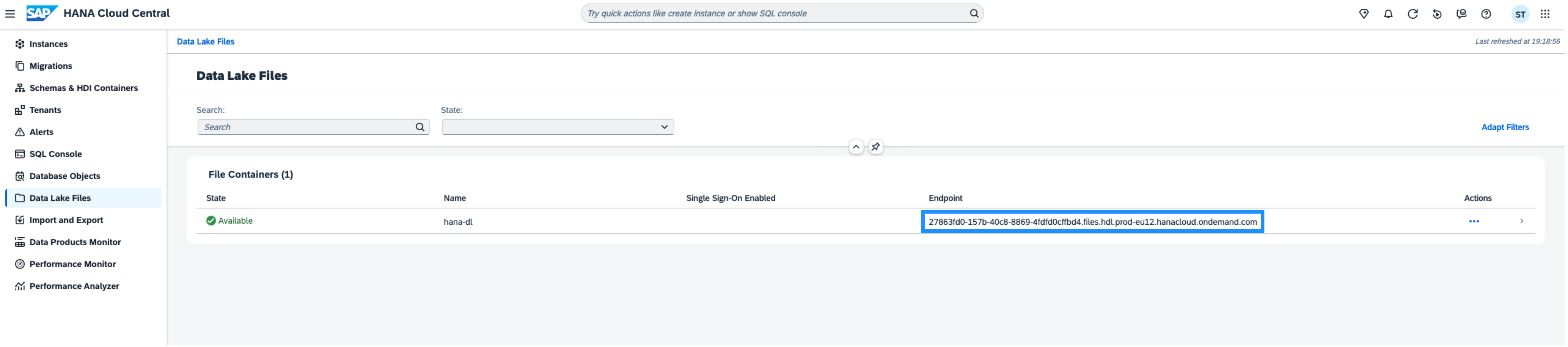
Access the instance and click **Manage Configurations**

Under the **Connections** section, select **Allow all IP addresses**



HDLFS Endpoint

You can find the HDLFS endpoint which is required for registered a data destination via the Data Lake Files tab

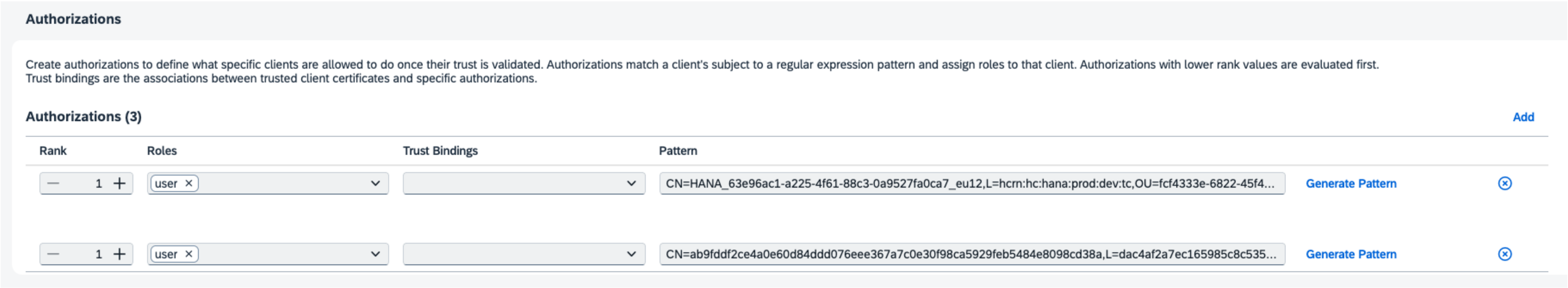


Important Note 1

When you first create a data destination to your HDLFS instance, you will get 201 created with the register and the mTLS subject patterns generated for this destination

Add these subject patterns to your HDLFS instance. Access the instance and click **Manage Configurations**.

Under the **Authorizations** section and add 2 new authorizations as follows. Each subject pattern will be 1 authorization.



Important Note 2

Before you can upload any new files to your HDLFS instance, you will need to generate a certificate to make a secure connection

You can follow along Step 3 and Step 4 in the developer guide to set this up